Road Pitch Estimation and Sensor Fusion in CARLA

Semantic Camera Measurements, MobileNet Regression, and Kalman Filtering

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EE4084 Project Video

Video Roadmap

Main project explanation

0:00	Introduction
0:20	Contents
0:45	Project goal
1:15	Final system overview
2:00	Results and discussion

Reproducibility and implementation

4:30	Tools used
5:30	Methods and theory
7:00	Code structure
8:15	Setup and running
9:30	Conclusion

The first part gives the project result and interpretation. Later sections explain how it was built and how it can be reproduced.

Project Goal

Problem

Road pitch affects vehicle longitudinal dynamics and autonomous driving state estimation.

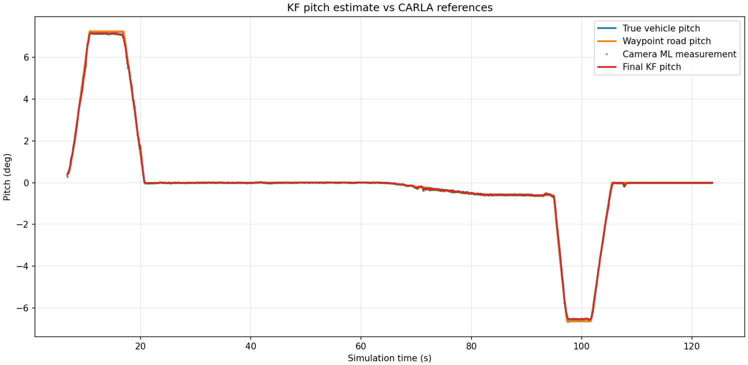
Goal

Estimate road pitch in CARLA using multiple simulated sensing sources and compare the results with CARLA reference signals.

Signals used

- Semantic camera
- IMU gyroscope
- GNSS baseline
- CARLA vehicle pitch
- CARLA waypoint pitch

Final Result First: Kalman Filter Output



Main observation

The Kalman Filter output follows the CARLA reference pitch closely during both uphill and downhill sections.

Error summary

Estimate	MAE	RMSE
Camera	0.028°	0.050°
KF	0.034°	0.063°
GNSS	0.023°	0.158°

What the Camera Model Sees

Semantic mask input

- Road pixels are retained
- Road-line pixels are retained
- Other objects are removed

Why this helps

The network sees only road geometry, so the pitch regression problem is simplified under controlled CARLA conditions.

Final System Overview

Camera branch

Semantic Camera → Road/Line Mask → MobileNetV3 → Camera Pitch

Filter branch

IMU Gyroscope → KF Prediction
Camera Pitch → KF Correction

Evaluation branch

GNSS → Baseline Comparison
CARLA Truth + Waypoint → Reference Signals

Key design choice

The original horizon/vanishing-point camera measurement was replaced with a learned pitch regressor.

Resulting filter

Since the camera model directly outputs pitch, the implemented fusion stage becomes a linear Kalman Filter.

Tools Used

Software

CARLA	0.9.16
Python	3.11
PyTorch	CUDA build
CUDA	13.2
OpenCV	image processing
MobileNetV3	pitch regression

Simulation setup

Map	Town05
Vehicle	Tesla Model 3 actor
Camera	Semantic segmentation
Sampling	20 Hz
Route	ramps + turns
GPU	RTX 5070 Ti

Full setup instructions and project files are intended to be available in the GitHub repository.

Dataset Generation

Collected data

- Semantic road masks
- True vehicle pitch
- Waypoint road pitch
- IMU and GNSS logs

Split

Training: 80%
Validation: 20%

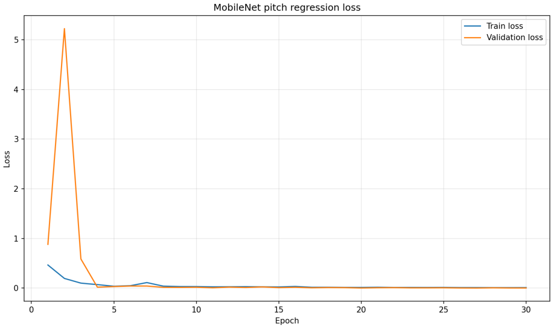
Dataset logic

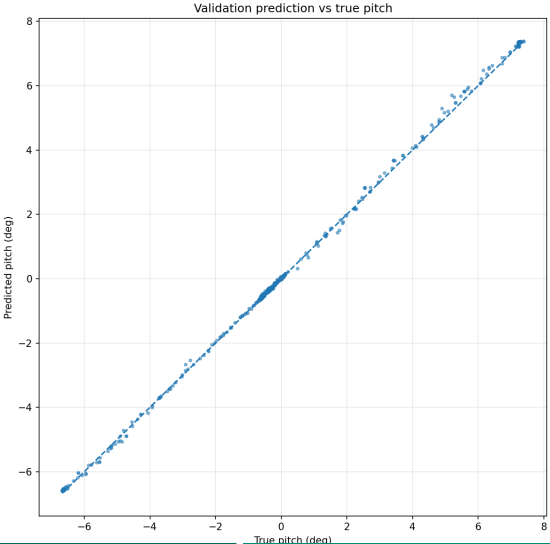
CARLA drives the selected route automatically. Each camera frame is converted into a road and road-line mask and paired with the corresponding pitch label.

Training target

The MobileNet model was trained using true vehicle pitch as the regression target.

MobileNet Training Results





Kalman Filter: Prediction Step

IMU-Based Pitch Propagation

The IMU gyroscope provides pitch rate measurements.

$$\hat{\theta}_k^- = \hat{\theta}_{k-1} + \omega_{y,k}\Delta t$$

Uncertainty Prediction

The uncertainty grows during prediction:

$$P_k^- = P_{k-1} + Q$$

Kalman Filter: Camera Correction

Camera Measurement

The MobileNet model produces a pitch estimate:

$$z_k = \theta_{camera}$$

Kalman Gain

$$K_k = \frac{P_k^-}{P_k^- + R}$$

State Correction

$$\hat{\theta}_k = \hat{\theta}_k^- + K_k \left(z_k - \hat{\theta}_k^- \right)$$

The camera estimate corrects accumulated IMU drift and improves long-term stability.

Kalman Filter: Parameters

Parameters

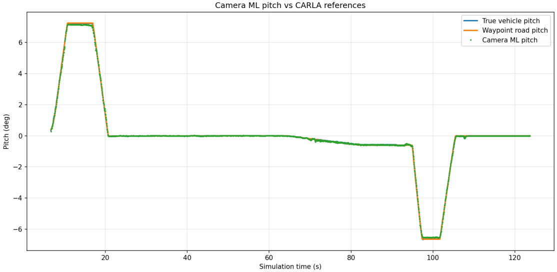
$$\omega_{y,k} = \text{IMU gyroscope pitch rate}$$

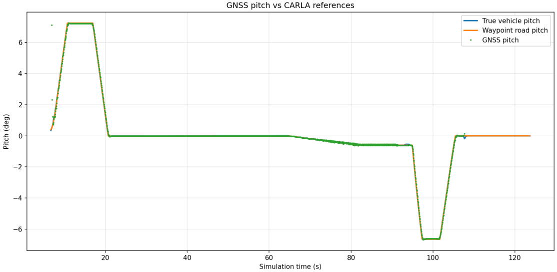
$$Q = 0.01$$

$$R = 0.01$$

The IMU provides short-term pitch propagation between camera measurements.

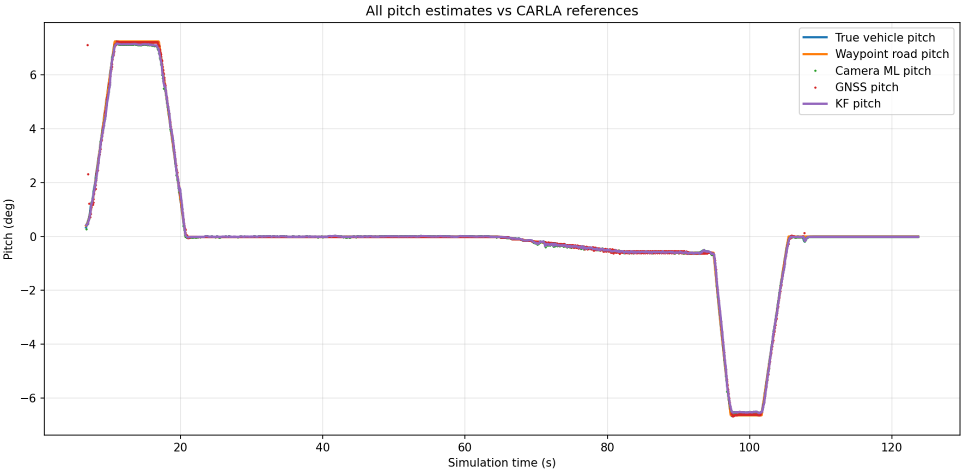
Camera and GNSS Results





GNSS pitch is useful as a baseline, but it depends on horizontal displacement and can become unavailable during stops.

Combined Results



Camera and KF remain available at each camera frame. GNSS is only evaluated when displacement is sufficient.

Quantitative Error Summary

Estimate	MAE (deg)	RMSE (deg)	Max (deg)
Camera vs true	0.028	0.050	0.408
Camera vs waypoint	0.032	0.057	0.465
GNSS vs true	0.023	0.158	6.611
GNSS vs waypoint	0.024	0.158	6.632
KF vs true	0.034	0.063	0.366
KF vs waypoint	0.035	0.062	0.349

Interpretation

The camera model is highly accurate in this controlled CARLA route. In less controlled conditions, IMU-assisted filtering is expected to become more useful when camera measurements degrade or disappear.

Code Structure

Main scripts

01	collect pitch dataset
02	train MobileNet model
05	run KF evaluation

Outputs

CSV logs, error summary, trained model, plots, mask images.

Run order

1. Start CARLA 0.9.16
2. Collect dataset
3. Train model
4. Run final KF script
5. Export plots and metrics

Setup and Running

Requirements

CARLA 0.9.16, Python 3.11, PyTorch CUDA build, OpenCV, torchvision, matplotlib.

Typical workflow

1. Download and run CARLA 0.9.16.
2. Create a Python 3.11 virtual environment.
3. Install dependencies listed in the GitHub repository.
4. Run dataset collection, training, and final evaluation scripts.

The public GitHub repository should contain exact commands, project structure, example outputs, and the final paper PDF.

Limitations and Conclusion

Limitations

- Controlled CARLA route
- Semantic masks instead of real RGB camera input
- Limited road/environment variety
- No real-world sensor validation yet

Conclusion

- CARLA pitch dataset generated
- MobileNet pitch regressor trained
- GNSS baseline evaluated
- IMU-camera KF implemented
- Results match CARLA references closely

Repository and paper links will be provided in the video description.

Sources and Thank You

Main References

Key sources cover monocular road estimation [1], the CARLA simulator [2], MobileNetV3 [3], and PyTorch [4].

- E. Ustunel and E. Masazade, "Iterative range and road parameters estimation using monocular camera on highways," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 1, pp. 139–151, 2023.
- A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An open urban driving simulator," in *Proceedings of the 1st Annual Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 78. PMLR, 2017, pp. 1–16. [Online]. Available: https://proceedings.mlr.press/v78/dosovitskiy17a.html
- A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, Q. V. Le, and H. Adam, "Searching for MobileNetV3," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 1314–1324.
- A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019. [Online]. Available: https://papers.nips.cc/paper/9015-pytorch-an-imperative-style-high-performance-deep-learning-library

Thank You for Watching